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Table of Content

1. Project Overview 5
    1.1 Introduction 5
    1.2 Project Approach 5
    1.3 Tools & Technologies 5
    1.4 Dashboard Pages 5
2. Business Objectives 6
    2.1 Market Structure 6
    2.2 Pricing 6
    2.3 Customer Engagement & Availability 6
    2.4 Policies & Operations 6
3. Business Problems 7
4. Project Goal 8
5. Dataset 8
    5.1 Dataset Overview 8
6. Data Cleaning & Preprocessing 9
    6.1 Libraries and Data Loading 9
    6.2 Column Standardization and Duplicate Removal 9
    6.3 Data Type Conversion 10
    6.4 Outlier Inspection 10
    6.5 Removing Low-Value Columns 11
    6.6 Country and Country Code 12
    6.7 Host and Listing Information 12
    6.8 Neighbourhood Data 13
    6.9 Validation of Minimum Nights and Availability 13
    6.10 Location and Booking Variables 13
    6.11 Pricing Variables 14
    6.12 Median Imputation 14

6.13 Review Status Feature 14
6.14 Final Review Data Cleaning 15
6.15 Exporting the Clean Dataset 16
6.16 Data Cleaning Principle 16
7. Exploratory Data Analysis Using Python 16
7.1 Purpose of the Analysis 16
7.2 Exploratory Analysis 17
7.3 Key Findings from EDA 17
8. Power BI Data Modeling 18
8.1 Data Model 18
8.2 Why Data Modeling Was Used 18
9. DAX Measures 19
9.1 Total Listings 19
9.2 Total Hosts 19
9.3 Listings With Reviews 19
9.4 Listings With Reviews % 20
9.5 Average Price 20
9.6 Average Minimum Nights 20
9.7 Average Rating 20
9.8 Average Service Fee 21
9.9 Verified Hosts % 21
9.10 Instant Bookable % 21
9.11 House Rules Enforcement Rates 22
9.12 Role of DAX in the Project 22
10. Dashboard Design & Business Insights 22
10.1 Executive Overview 23
10.2 Pricing Analysis 24
10.3 Reviews & Availability 25
10.4 House Rules & Policies 25

10.5 Neighbourhood Group Drill-Through 26
11. Key Business Findings 27
12. Business Recommendations 28
12.1 Improve Pricing Analysis 28
12.2 Investigate High-Availability Segments 28
12.3 Improve Customer Engagement 29
12.4 Evaluate Minimum-Stay Policies 29
13. Analytical Limitations 29
14. Conclusion 30
15. Future Improvements 30

1. Project Overview

1.1 Introduction

This project analyzes Airbnb listing data to understand the characteristics, pricing patterns, availability, review activity, and policies of Airbnb listings.

The project combines **Python for data cleaning and exploratory analysis** with **Power BI for interactive business intelligence and visualization**.

The final deliverable is an interactive Power BI dashboard designed to transform raw Airbnb listing data into meaningful business insights that can support better understanding of the Airbnb marketplace.

1.2 Project Approach

The project follows an end-to-end data analytics workflow:

• Data Extraction & Cleaning: Python (Pandas) was used to handle missing values, type conversion, regex pattern extraction, and distribution-based imputation.

• Data Modeling & DAX: Cleaned data was imported into Power BI, organized into a relational model, and enhanced with dynamic DAX metrics.

• Visualization & Insights: Interactive multi-page dashboards were designed to present business findings and support decision-making.

1.3 Tools & Technologies

• Python (Pandas, NumPy, Matplotlib): Data cleaning, transformation, regex feature extraction, and exploratory analysis.

• Power BI Desktop: Relational data modeling, DAX measure creation, and interactive dashboard design.

• Excel / CSV: Intermediary storage for raw and cleaned datasets.

1.4 Dashboard Pages

The Power BI report contains five dedicated analytical pages:

1. Executive Overview

2. Pricing Analysis

3. Reviews & Availability

4. House Rules & Policies

5. Neighbourhood Group Drill-Through

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2. Business Objectives

The analysis focuses on four main areas:

2.1 Market Structure

- Analyze how listings are distributed across neighbourhood groups.
- Understand the distribution of different room types.

2.2 Pricing

- Analyze average listing prices and service fees.
- Compare pricing across neighbourhood groups and room types.
- Examine minimum-stay requirements.

2.3 Customer Engagement & Availability

- Measure the percentage of listings with reviews.
- Compare review coverage across room types.
- Analyze listing availability.
- Examine Instant Booking patterns.

2.4 Policies & Operations

- Analyze cancellation policy distribution.
- Identify the most common house rules.
- Compare minimum-stay requirements across neighbourhoods.

3. Business Problems

The project addresses the following business questions:

Problem 1 — Market Concentration

Where is Airbnb listing supply concentrated?

Understanding supply concentration helps identify the largest and most competitive markets.

Problem 2 — Pricing Differences

Does neighbourhood have a meaningful impact on listing prices?

The analysis investigates whether location alone can explain pricing differences or whether other listing characteristics should be considered.

Problem 3 — Customer Engagement

How does review activity differ across room types?

Review coverage is used as an indicator of customer engagement and listing activity.

Problem 4 — Availability & Booking

Do some room types show substantially higher availability or different booking behavior?

This helps identify segments that may require further investigation of inventory utilization and demand.

Problem 5 — Operational Policies

How do minimum stays, cancellation policies, and house rules vary across the marketplace?

Understanding these differences can help identify operational patterns and potential areas for optimization.

4. Project Goal

The overall goal of the project is to:

Identify meaningful patterns, business problems, and opportunities within Airbnb listing data and transform them into insights that can support better pricing, inventory, and operational decisions.

5. Dataset

5.1 Dataset Overview

The project uses an Airbnb listing dataset containing information about properties, hosts, locations, pricing, reviews, availability, booking preferences, and operational policies.

The dataset includes listing-level records and was prepared before being imported into Power BI for further analysis.

Main Data Categories

Category	Examples
Listing Information	Listing ID, Name, Room Type
Host Information	Host ID, Host Name, Host Verification
Location	Neighbourhood Group, Neighbourhood, Latitude, Longitude
Pricing	Price, Service Fee, Minimum Nights
Reviews	Number of Reviews, Last Review, Review Rate
Availability	Availability 365, Instant Bookable

Policies	Cancellation Policy, House Rules

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6. Data Cleaning & Preprocessing

6.1 Libraries and Data Loading

Python was used as the main data preparation tool.

The following libraries were used:

```
import pandas as pd

import numpy as np
```

The dataset was loaded using Pandas:

```
data0 = pd.read_csv('Airbnb_Open_Data.csv')
```

Initial inspection was performed using:

```
data0.head()

data0.shape

data0.info()
```

These steps were used to understand the dataset structure, dimensions, column types, and initial data quality.

6.2 Column Standardization and Duplicate Removal

Column names were standardized by removing unnecessary spaces:

```
data.columns = data.columns.str.strip()
```

Duplicate records were identified and removed:

```
data.duplicated().sum()

data.drop_duplicates(inplace=True)
```

This ensured that duplicate records would not affect subsequent calculations and analysis.

6.3 Data Type Conversion

The last review column was converted into a proper datetime format:

```
data["last review"] = pd.to_datetime(data["last review"])
```

The price and service fee columns contained currency symbols and commas. These were removed before converting the columns to numeric values:

```
data["price"] = (
    data["price"]
      .replace('[\$,]', '', regex=True)
      .astype(float)
)


data["service fee"] = (
    data["service fee"]
      .replace('[\$,]', '', regex=True)
      .astype(float)
)
```

Descriptive statistics were then generated to inspect the resulting numerical variables.

6.4 Outlier Inspection

Boxplots were used to visually inspect price and service fee:

```
import matplotlib.pyplot as plt


plt.boxplot([data['price'], data['service fee']])

plt.show()
```

Based on the visual inspection performed during the project, no outliers were identified in these variables.

6.5 Removing Low-Value Columns

The license column was removed because it contained an extremely high proportion of missing values and did not provide sufficient analytical value for the project:

```
# License column removal

data.drop(columns=["license"], inplace=True)

# Extract house rules into a normalized relational table

house_rules_table = data.loc[data["house_rules"].notna(), ["id",
"house_rules"]].reset_index(drop=True)

data.drop(columns=["house_rules"], inplace=True)

# Feature extraction using Regex

house_rules_table["No Smoking"] =
house_rules_table["house_rules"].str.contains("no smoking|smoking not
allowed|non-smoking", case=False, regex=True, na=False)

house_rules_table["No Pets"] =
house_rules_table["house_rules"].str.contains("no pet|pets not allowed|no
animals", case=False, regex=True, na=False)
```

house_rules_table["No Parties"] = house_rules_table["house_rules"].str.contains("no party|no parties", case=False, regex=True, na=False)

house_rules_table["Quiet Hours"] = house_rules_table["house_rules"].str.contains("quiet", case=False, regex=True, na=False)

This allowed analyzing house rules in Power BI without keeping raw text in the main dataset.

6.6 Country and Country Code

The country and country code columns were inspected before handling their missing values.

Missing country values were replaced with:

data["country"] = data["country"].fillna("United States")

Missing country codes were replaced with:

data["country code"] = data["country code"].fillna("US")

This was done because the dataset is focused on Airbnb listings in the United States.

6.7 Host and Listing Information

Missing values in host verification were imputed based on the existing class distribution rather than dropping records:

probs = data["host_identity_verified"].value_counts(normalize=True)

data.loc[missing_mask, "host_identity_verified"] = np.random.choice(probs.index, size=missing_mask.sum(), p=probs.values)

Missing host names and listing names were imputed using relational mapping based on matching host IDs and coordinates, while remaining unmapped records were labeled as "Unknown Host" or "Unknown".

6.8 Neighbourhood Data

The neighbourhood group values were checked for inconsistent spelling.

Two spelling errors were corrected:

```
data["neighbourhood group"] = data["neighbourhood group"].replace({
 "brookln": "Brooklyn",
 "manhatan": "Manhattan"})
```

Missing values in neighbourhood and neighbourhood group were imputed using their hierarchical relationships and group-level distributions to retain the geographic structure across all listings.

6.9 Validation of Minimum Nights and Availability

The minimum nights variable was checked for invalid values.

Only values between 1 and 365 were retained:

```
# Retain valid ranges for minimum nights and availability
data = data[(data["minimum nights"] > 0) & (data["minimum nights"] <= 365)]
data = data[(data["availability 365"] >= 0) & (data["availability 365"] <= 365)]
```

This prevented impossible values from affecting the analysis.

6.10 Location and Booking Variables

Missing geographic coordinates (lat, long) were imputed using the mean coordinates of each neighbourhood to preserve spatial consistency:

data.loc[mask, "lat"] = data.loc[mask, "neighbourhood"].map(lat_mean)

data.loc[mask, "long"] = data.loc[mask, "neighbourhood"].map(long_mean)

Missing values for instant_bookable and cancellation_policy were imputed using probability distribution sampling based on overall observed ratios.

6.11 Pricing Variables

Listings with missing price or service fee values were removed:

data = data.dropna(subset=["price"])

data = data.dropna(subset=["service fee"])

This ensured that pricing KPIs and comparisons were calculated only from records containing valid pricing information.

6.12 Median Imputation

Missing values in Construction year were replaced using the median:

Median imputation for continuous attributes

data["Construction year"] = data["Construction year"].fillna(data["Construction year"].median())

data["calculated host listings count"] = data["calculated host listings count"].fillna(data["calculated host listings count"].median())

Median imputation was used to retain records while reducing the influence of extreme values compared with mean-based imputation.

6.13 Review Status Feature

Review-related columns were examined together:

data[

```
    ['number of reviews',
     'last review',
     'reviews per month',
     'review rate number']
].isna().value_counts()
```

A new feature called review_status was created:

```
data['review_status'] = np.where(
    data['last review'].isna() &
    data['reviews per month'].isna(),
    'No Reviews',
    'Has Reviews'
)
```

This feature classifies listings into two groups:

- **Has Reviews**
- **No Reviews**

The feature was created because the missing-value patterns in last review and reviews per month provided an indicator of whether review activity existed.

6.14 Final Review Data Cleaning

Review-related missing values were handled using logical relationships. Listings without review dates were assigned a review count of zero, while active listing gaps were imputed using median values and observed rating distributions:

```
data.loc[mask_no_reviews, "reviews per month"] = 0

data.loc[mask_no_reviews, "number of reviews"] = 0
```

6.15 Exporting the Clean Dataset

After completing the preprocessing steps, the cleaned dataset was exported as a Excel file:

```
data.to_excel("airbnb_cleaned.xlsx", index=False)
```

The resulting cleaned dataset was then used as the basis for the Power BI analysis.

6.16 Data Cleaning Principle

The overall cleaning approach followed one main principle:

A data-cleaning decision should be based on data quality, analytical importance, and business relevance rather than simply removing every record containing a missing value.

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7. Exploratory Data Analysis Using Python

7.1 Purpose of the Analysis

After cleaning the dataset, exploratory data analysis (EDA) was performed using Python to understand the main characteristics and patterns within the Airbnb data before building the Power BI dashboard.

The analysis focused on:

- Listing distribution
- Room type distribution
- Pricing patterns
- Review activity
- Availability

- Minimum-night requirements
- Relationships between selected variables

The purpose of this stage was to identify important patterns and determine which findings could provide business value.

7.2 Exploratory Analysis

The cleaned dataset was explored using statistical summaries and visualizations.

Key variables were analyzed to understand their:

- Distribution
- Central tendency
- Variation
- Missing-value patterns
- Relationships with other business variables

Particular attention was given to **price, service fee, minimum nights, number of reviews, availability, room type, and neighbourhood group**, as these variables are directly relevant to the business questions defined for the project.

7.3 Key Findings from EDA

The exploratory analysis helped identify several patterns that were later investigated in Power BI:

- Listing supply is concentrated in specific neighbourhood groups.
- Entire Homes and Private Rooms represent the majority of the available inventory.
- Average prices are relatively similar across neighbourhood groups.
- Review activity differs across room types.

- Availability varies considerably between room types.
- Minimum-stay requirements differ across neighbourhood groups.

These findings were used to guide the design of the Power BI dashboard and the selection of the most relevant KPIs and visualizations.

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8. Power BI Data Modeling

8.1 Data Model

The cleaned dataset was imported into Power BI and organized into a structured analytical model.

The model includes:

- **airbnb_cleaned** — Main listing dataset
- **Calendar** — Used for date-related analysis
- **house_rules_table** — Used for house rules analysis
- **Measures_table** — Used to organize DAX measures

The model was designed to support interactive filtering and reusable calculations across the dashboard.

8.2 Why Data Modeling Was Used

Data modeling provides a structured foundation for the dashboard by allowing:

- Consistent calculations across visuals
- Reusable measures
- Interactive filtering
- Separation of data and calculations
- More maintainable dashboard development

This ensures that the dashboard is not based on manually calculated or static values.

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9. DAX Measures

DAX was used to create dynamic KPIs and analytical measures within Power BI.

9.1 Total Listings

Total Listings =

DISTINCTCOUNT(airbnb_cleaned[id])

This measure calculates the number of unique Airbnb listings.

9.2 Total Hosts

Total Hosts =

DISTINCTCOUNT(airbnb_cleaned[host id])

This measure calculates the number of unique hosts represented in the dataset.

9.3 Listings With Reviews

Listings With Reviews =

CALCULATE(

[Total Listings],

airbnb_cleaned[review_status] = "Has Reviews"

)

This measure counts listings that have review activity.

9.4 Listings With Reviews %

```
Listings With Reviews % =
DIVIDE(
    [Listings With Reviews],
    [Total Listings],
    0
)
```

This measure calculates the percentage of listings that have reviews.

It is particularly useful for comparing review coverage across different room types and neighbourhood groups.

9.5 Average Price

```
Average Price =
AVERAGE(airbnb_cleaned[price])
```

This measure calculates the average listing price and is used throughout the pricing analysis.

9.6 Average Minimum Nights

```
Average Minimum Nights =
AVERAGE(airbnb_cleaned[minimum nights])
```

This measure calculates the average minimum stay requirement.

9.7 Average Rating

```
Average Rating ⭐ =
```

AVERAGE(airbnb_cleaned[review rate number])

This measure calculates the overall average guest rating score across listings.

9.8 Average Service Fee

Average Service Fee =

AVERAGE(airbnb_cleaned[service fee])

This measure calculates the average service fee charged per listing.

9.9 Verified Hosts %

Verified Hosts % =

DIVIDE(

CALCULATE([Total Listings], airbnb_cleaned[host_identity_verified] = "verified"),

[Total Listings],0)

This measure calculates the percentage of listings managed by identity-verified hosts.

9.10 Instant Bookable %

Instant Bookable % =

DIVIDE(

CALCULATE([Total Listings], airbnb_cleaned[instant_bookable] = "TRUE"),

[Total Listings],

0

)

This measure calculates the proportion of listings that allow immediate booking without manual host approval.

9.11 House Rules Enforcement Rates

```
No Smoking % =
DIVIDE(
   CALCULATE(COUNT(house_rules_table[id]), house_rules_table[No Smoking] = TRUE()),
   COUNT(house_rules_table[id]),
   0
)
```

Similar DAX measures were created for No Pets %, No Parties %, and Quiet Hours % to calculate rule compliance percentages across normalized house rules data.

9.12 Role of DAX in the Project

DAX measures were used instead of relying only on static calculations because they dynamically respond to filters and selections made by the user.

For example, when a specific neighbourhood or room type is selected, the KPI values and visualizations update according to the selected context.

This makes the dashboard suitable for interactive business analysis rather than simple static reporting.

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10. Dashboard Design & Business Insights

The Power BI dashboard was designed to provide an interactive view of the Airbnb marketplace and translate analytical results into business insights.

The dashboard is divided into several analytical pages, each addressing a specific business area.

10.1 Executive Overview

The Executive Overview provides a high-level summary of the marketplace.

Main Areas Analyzed

- Listing distribution by neighbourhood
- Listing distribution by room type
- Review coverage by room type

Key Business Insights

Supply Concentration

Listings are highly concentrated in Manhattan and Brooklyn, indicating that Airbnb supply is concentrated in a limited number of markets.

Room Type Structure

Entire Homes and Private Rooms dominate the marketplace, while Shared Rooms and Hotel Rooms represent a much smaller portion of the inventory.

Review Engagement

Entire Home listings have the highest review coverage at 100%, while Hotel Room listings have the lowest at 77%.

Business Value

The Executive Overview allows decision-makers to quickly understand the overall structure of the marketplace and identify segments that require deeper investigation.

10.2 Pricing Analysis

The Pricing Analysis page focuses on price, service fees, minimum-night requirements, and the relationship between price and availability.

Key Business Insights

Neighbourhood Pricing

Average listing prices are highly consistent across neighbourhood groups, ranging approximately from $624 to $629.

Business Implication

Neighbourhood alone does not appear to be a strong pricing differentiator in this dataset, suggesting that other factors should be considered when evaluating pricing.

Neighbourhood & Room Type

Although overall neighbourhood prices are relatively stable, meaningful differences appear when specific room types are compared across neighbourhoods.

Business Implication

Pricing analysis should consider both neighbourhood and room type rather than relying on neighbourhood averages alone.

Price & Availability

Hotel rooms have the highest average price at approximately $663 and the highest average availability at approximately 220 days.

Business Implication

The hotel segment should be investigated further to determine whether its high availability reflects excess inventory, different demand patterns, or a different operating model.

10.3 Reviews & Availability

This page focuses on customer engagement, listing availability, and booking behavior.

Key Business Insights

Review Coverage

Approximately 85% of listings have reviews, indicating strong customer engagement across the marketplace.

Availability

Hotel and Shared Room listings show substantially higher availability than Private Rooms and Entire Homes.

Instant Booking

Instant Booking is almost evenly split across listings, indicating that hosts are divided between immediate booking and host-confirmation models.

Business Value

These analyses help identify differences in customer engagement, inventory availability, and booking behavior across listing segments.

10.4 House Rules & Policies

This page examines operational policies that can affect the guest experience and booking flexibility.

Key Business Insights

Cancellation Policies

Flexible, Moderate, and Strict cancellation policies are relatively evenly distributed across the marketplace.

Minimum Nights

Manhattan has the highest average minimum-stay requirement, while other neighbourhoods have noticeably lower requirements.

House Rules

No Smoking is the most common enforced house rule, followed by No Pets, Quiet Hours, and No Parties.

Business Value

The analysis highlights differences in operational policies and identifies areas where minimum-stay requirements and house rules may affect customer flexibility and the overall booking experience.

10.5 Neighbourhood Group Drill-Through

A drill-through page was created to allow users to move from the overall dashboard into a more detailed analysis of a selected neighbourhood group.

The page provides additional information about:

- Total Listings
- Average Price
- Average Availability
- Average Rating
- Room Type Distribution
- Service Fees
- Instant Booking
- Detailed listing segments

Business Value

The drill-through functionality allows users to investigate a selected neighbourhood at a more granular level without overcrowding the main dashboard pages.

It supports questions such as:

How does this neighbourhood differ across room types, pricing, availability, and booking characteristics?

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11. Key Business Findings

The analysis identified six major findings:

1. Supply Concentration

Manhattan and Brooklyn represent the largest concentrations of Airbnb listing supply.

2. Neighbourhood Alone Does Not Explain Pricing

Average prices are remarkably similar across neighbourhood groups, while room type introduces more meaningful differences.

3. Availability Differs by Room Type

Hotel and Shared Room listings show substantially higher availability than Private Rooms and Entire Homes.

4. Strong Review Coverage

Approximately 85% of listings have reviews, indicating broad customer engagement across the marketplace.

5. Minimum-Stay Differences

Manhattan has the highest average minimum-stay requirement.

6. No Dominant Cancellation Strategy

Flexible, Moderate, and Strict cancellation policies have relatively similar distributions.

---

12. Business Recommendations

Based on the identified patterns, several areas can be considered for further business action.

12.1 Improve Pricing Analysis

Pricing decisions should not rely on neighbourhood alone.

Additional factors such as:

- Room Type
- Availability
- Reviews
- Property Characteristics
- Demand

should be considered when evaluating pricing strategies.

12.2 Investigate High-Availability Segments

Hotel and Shared Room listings show higher availability.

Further analysis should determine whether this reflects:

- Excess inventory
- Different demand patterns
- Different operating models

before making pricing or inventory decisions.

12.3 Improve Customer Engagement

Hotel Room listings have the lowest review coverage among the analyzed room types.

This segment could be investigated further to understand the reasons behind lower review engagement and identify opportunities to encourage more guest feedback.

12.4 Evaluate Minimum-Stay Policies

Higher minimum-stay requirements, particularly in Manhattan, should be evaluated against availability and customer engagement to understand their potential effect on booking flexibility.

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13. Analytical Limitations

The findings should be interpreted within the limitations of the available data.

- The **Year** variable is based on Last Review, not listing creation date.
- Actual revenue is not directly available.
- Occupancy data is not directly available.
- Listing price does not necessarily represent the final transaction price.
- Availability does not directly measure demand.
- Observed relationships do not necessarily imply causation.

Therefore, the dashboard identifies patterns and potential opportunities but cannot independently confirm causal business relationships.

---

14. Conclusion

This project transformed raw Airbnb listing data into an interactive Business Intelligence dashboard using Python and Power BI.

The analysis provided insights into:

- Marketplace supply concentration
- Room type distribution
- Pricing patterns
- Review engagement
- Availability
- Booking behavior
- Operational policies

The results show that **neighbourhood alone is not sufficient to explain pricing differences**, while room type and other listing characteristics provide additional context.

The dashboard therefore serves not only as a reporting tool, but also as an analytical framework for identifying areas that require further investigation and potential business optimization.

---

15. Future Improvements

The analysis could be strengthened by integrating additional business data such as:

- Actual booking data
- Occupancy rates
- Revenue
- Booking frequency

- Seasonal demand
- Customer demographics

Combining these variables with the current listing-level data would allow deeper analysis of demand, revenue performance, and pricing effectiveness.